CTF Workshop 1: Introduction!

What exactly is a CTF?

- A CTF (Capture The Flag) is a cybersecurity competition where players solve challenges to find flags, which are usually strings of text
  - Examples: gcssCTF{flag-example1}, GCSS{flag_3x4mp1e}

- Some examples of things that you actually do in these challenges:
  - Analyze files, websites, code, or networks
  - Exploit known vulnerabilities
  - Decode hidden information

- 2 types of CTFs: Jeopardy style (most common) and Attack-Defense style
  - **Jeopardy style**: challenges are open to everyone. Amount of points for a challenge decreases depending on how many solves.
  - **Attack-Defense**: teams run and defend their own vulnerable systems. Points are earned by keeping your system up as well as stealing flags from other teams.

CTF Categories

- Jeopardy style CTF challenges come in a few different categories:
  - **Crypto**: extract hidden messages in ciphers, encodings and break weak encryption
  - **Digital Forensics**: extract hidden data in files and images
  - **Web**: exploit vulnerabilities in web apps, such as SQL injection or authentication bypass
  - **Binary Exploitation (pwn)**: exploit low level program vulnerabilities such as buffer overflows
  - **Reverse Engineering**: convert or translate a program back into its source code

Crypto(graphy)



- Crypto challenges often involve breaking misused/weak cryptography (for example small primes in RSA), not real world encryption
- Common challenges use classical ciphers (Caesar, Vigenere, Substitution), encodings (Base64, Hex, Binary) or simple operations (XOR, rotation, shifting)
- **CyberChef & dCode** will be your best friends for crypto challenges

Quick Crypto Demo!

It’s your turn!

Visit the GCSS GitHub, navigate to the CTF-Workshop1 repo and attempt the challenge “intro”.

github.com/guelphcss

It’s your turn!

That was a warmup, easy enough right?
Now, attempt the challenge “Easy1”.

github.com/guelphcss

Hint: what encoding ends in = ?

It’s your turn!

Okay last one, try the challenge “hashcrack”. The link will take you to picoGym. Make an account and attempt the challenge!

github.com/guelphcss

Hint: Try to identify what hash is being used

Hint: what is hashcat and how does it work?

Tips for crypto challenges

- Look for patterns and repetition in ciphertexts, sometimes can indicate what kind of cipher you're working with.
- Identify the type of cipher/encoding.
- Challenge name can sometimes be a hint
- Google anything you don't know (not cheating)

Digital Forensics

- Digital Forensics challenges involve investigating files and digital “artifacts”
- Common forensics challenges can include investigating files with hidden data (images, PDFs), steganography (data hidden inside an image), network captures (pcap files) and more.
- Some common forensics tools: **file, strings, exiftool, wireshark, binwalk, steghide**



Quick Forensics Demo!

It’s your turn!

Attempt the first forensics challenge called “checkerboard”

github.com/guelphcss

It’s your turn!

Next, try “logo”. This one’s a bit trickier.

github.com/guelphcss

Hint:



It’s your turn!

Try the last challenge “baby-exfil”. This one’s pretty tough.

github.com/guelphcss

Tips for forensics challenges

- Don’t trust appearances, files are not always what they seem
- Identify file type, change approach based on this
- Always inspect files (metadata, strings, etc), look for places where data can be hidden
- Many tools available, some are better than others for certain situations, create a list of tools that you find to be more commonly used

Thank you for coming!

Don’t miss CTF Workshop 2!

More info will be posted closer to the event date!